

Collision Frequency Trends for Various Vehicle Types in the Greater Toronto Area

INF2167 Project - Group 7

Stella Gregorski Yiyang Qin

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Introduction

Why is traffic system important?

- Economic and social impacts
- Urban areas face increasing collision risks
- Data-driven analysis supports evidence-based planning

Motivation

What temporal and spatial patterns exist in Toronto traffic collisions, and how do collision risks vary across different travel modes?

- Study on Wujiang, China ([Cheng et al. 2018](#)) discovered high-risk spot at the connection with a major city, which is associated with high traffic volumes, ongoing construction projects, and high proportion of large trucks.
- Scientists in Iran ([Yousefzadeh-Chabok et al. 2016](#)) conducted a time series analysis in Zanjan Province for predicting the future morality in traffic, and the models shows a descending trend in traffic morality, associated with stricter traffic enforcement and education as time passed.
- Investigation on Toronto collision patterns supports evidence-based transportation planning and resource allocation

Data & Methods

Data Source & Structure

- The Toronto Police Service maintains data on traffic collisions they are called to respond to
- This data includes attributes such as:
 - **Month, Weekday, and Hour** when the collision took place
 - **Neighbourhood** where the collision took place
 - Binaries for whether five different types of vehicle were involved in the collision
 - These include **Automobile, Passenger Vehicle (public transportation), Motorcycle, Bicycle, and Pedestrian**
- Data collected between 2020 and 2026 was sourced from (Toronto Police Service 2023)

Data Cleaning & Pre-Processing

- R (R Core Team 2026) and the janitor package (Firke et al. 2024) were used to clean the data and optimize it for analysis
- Several columns were renamed using more semantic language
 - e.g. *"FTR_COLLISIONS"* renamed to *"FAILURE_TO_REMAIN"*
- All-capitals column names standardized to snake_case using janitor's clean_names() function
- Columns added for month_number (1-12, January - December) and dow_number (1-7, Sunday - Saturday) to simplify ordered temporal visualizations

Methods: Exploratory Data Analysis

This project used the R programming language (R Core Team 2026) and the tidyverse library (Wickham et al. 2019) to do reproducible data analysis. High level steps included:

- Looking at the temporal, spatial, and vehicle type columns using `unique()` to check that all values make sense for the type of data being reported
 - e.g. making sure that all values in the `occ_month` column contain month names, spelled correctly and consistently stylized
- Using `group_by()` and `mutate()` to calculate the percentage of total collisions that happened in a given time frame or neighbourhood
- Using the `pivot_longer()` function to melt the vehicle type columns from the original dataframe into one column containing vehicle type (`vehicle_type`) and another containing the binary cell values (`involved_yesno`)
 - This long dataframe was then filtered to only rows with "YES" in the `involved_yesno` column
 - This changes the observational unit from individual collisions to individual vehicles involved in collisions, with five rows per collision

Methods: Data Visualization

Statistical visuals and summary tables were generated using the *ggplot2* library from (Wickham et al. 2019), the patchwork library (Pedersen 2025) for concatenated figures, and kableExtra (Zhu [aut et al. 2026) for clean tables.

- Line graphs help to understand temporal trends in collision frequency
- Summary tables help to understand the most prominent spatial and temporal hotspots

Methods: Statistical Modeling

- Linear regression of the form $y = \beta_0 + \beta_1 x_1 + \dots + \beta_n x_n + \varepsilon$ was used to better understand the relationship between variables
- Temporal, spatial, and vehicle type variables were used as predictors, with collision frequency as the goal outcome variable
- This took the following form in R syntax:

```
1 model <- lm(y ~ x_1 + ... + x_n, data = tps_collisions)
```

Results

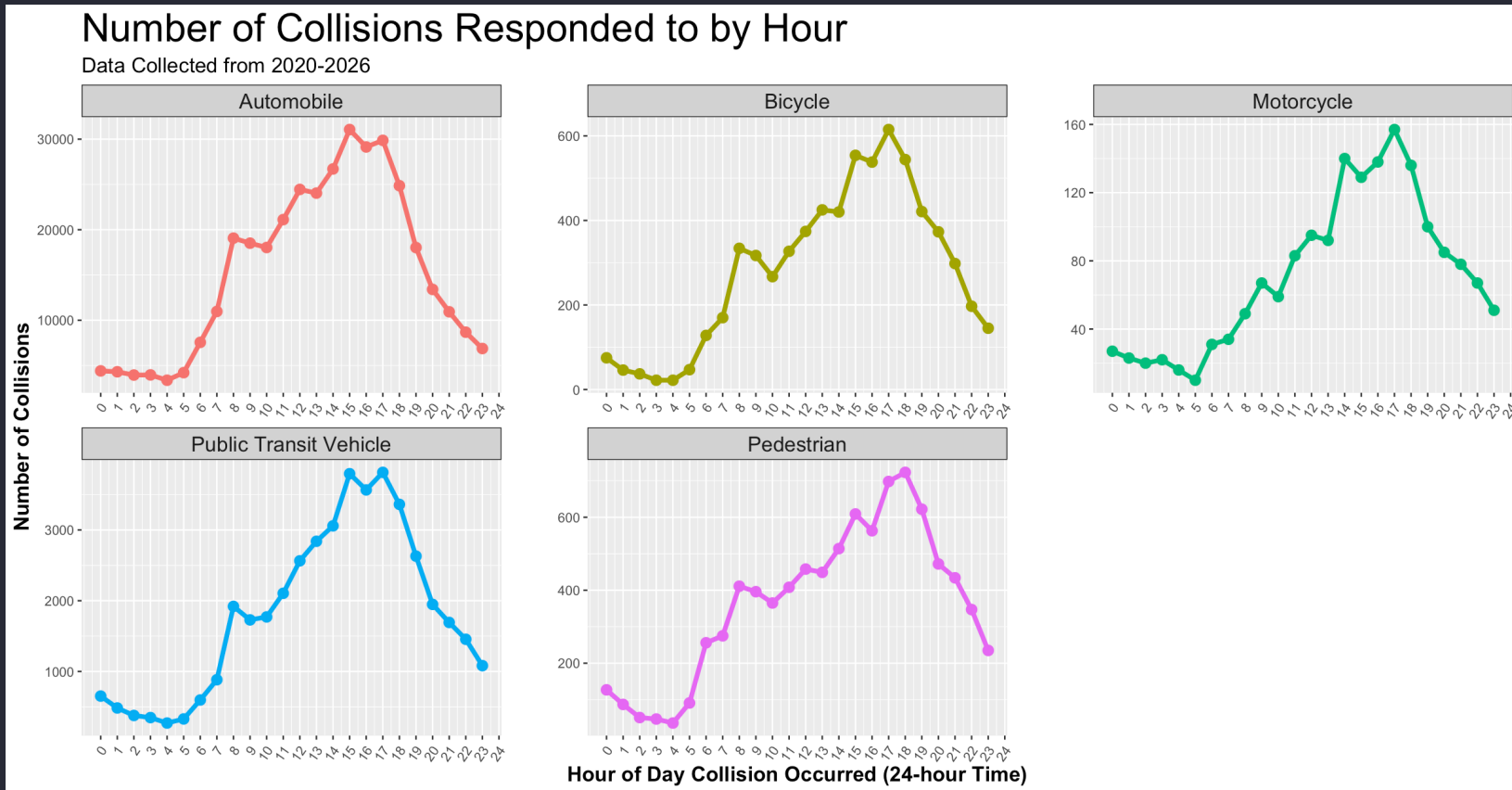
Spatial & Summary Trends

- Wexford/Maryvale is a spatial hotspot for collisions
- Temporal variation observed in collision frequency for different vehicle types

Vehicle Type	Hour with Highest Number of Recorded Collisions	Day of Week with Highest Number of Recorded Collisions	Month with Highest Number of Recorded Collisions	Neighbourhood with Highest Number of Recorded Collisions
Automobile	15	Friday	February	Wexford/Maryvale (119)
Bicycle	17	Wednesday	July	Annex (95)
Motorcycle	17	Saturday	August	Wexford/Maryvale (119)
Public Transit Vehicle	17	Saturday	January	Wexford/Maryvale (119)
Pedestrian	18	Wednesday	November	Wexford/Maryvale (119)

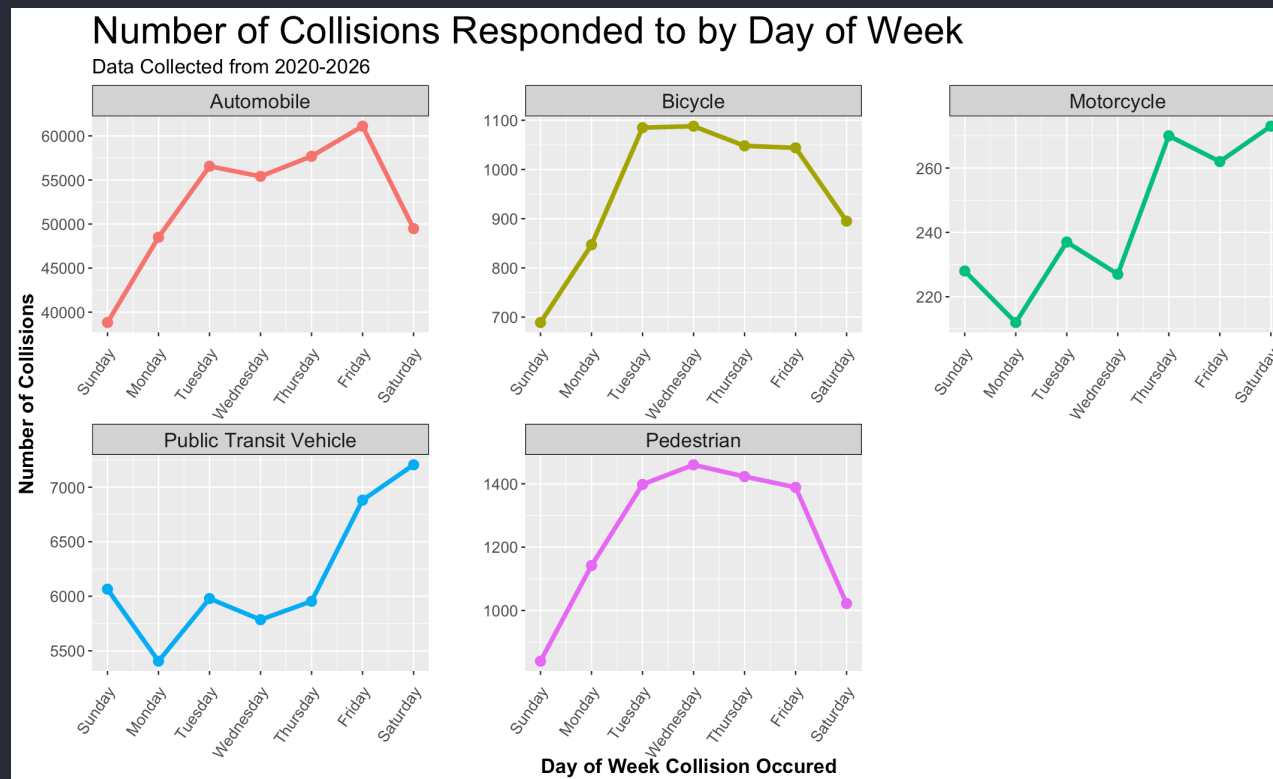
Hourly Trends

- Similar across all vehicle types
- Overnight collisions are low, with a sharp increase between **05:00** and **09:00**
- Daily peak between **15:00** and **18:00**



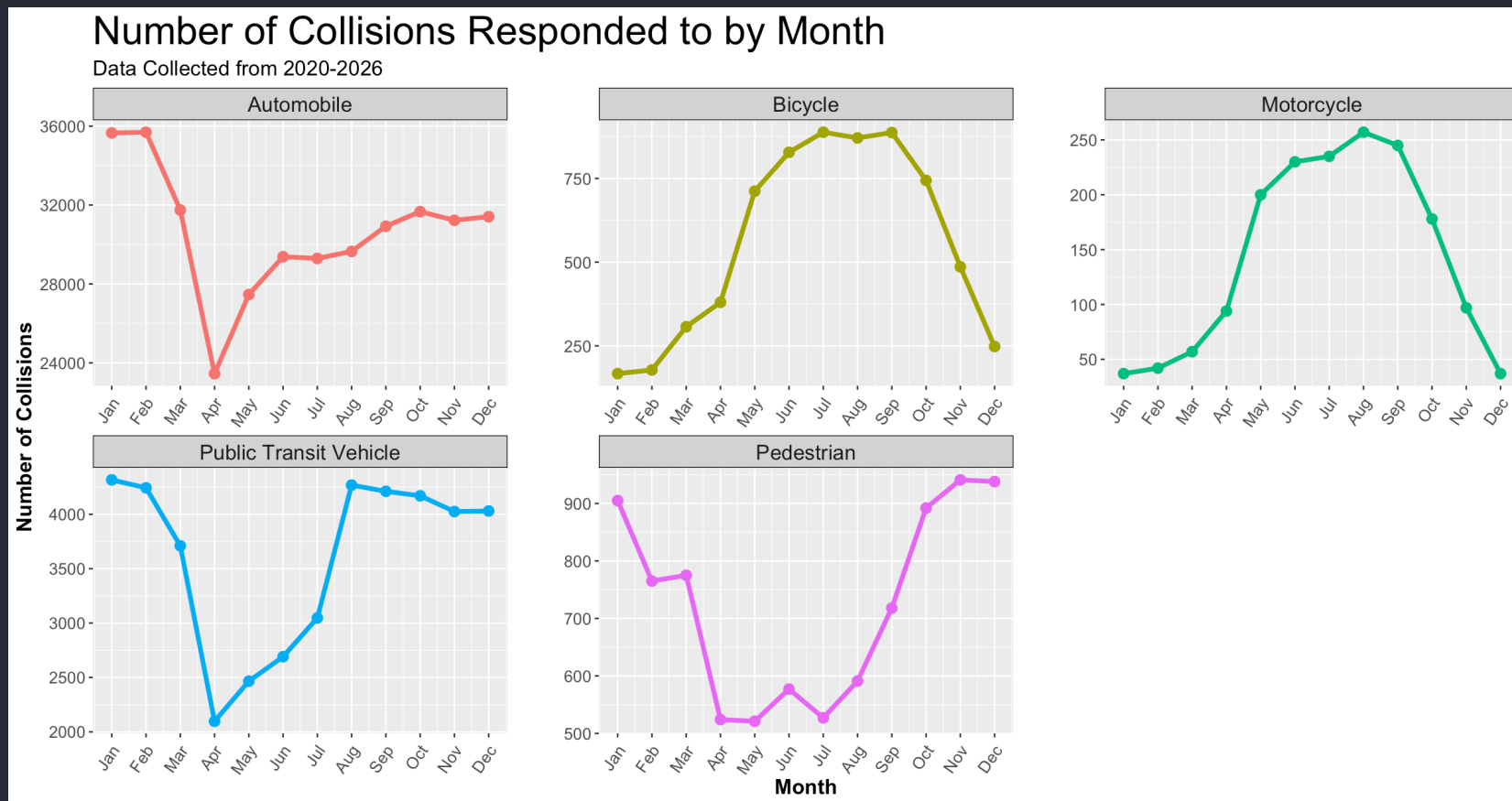
Weekly Trends

- **Bicycles** and **pedestrians** experience peak collisions in the middle of the week
- **Automobiles** experience a steady increase throughout the week, a sharp peak on Friday, and then a sharp decrease during the weekend
- Friday and Saturday are peak days for both **motorcycles** and **public transit**



Monthly Trends

- For **bicycles** and **motorcycles**, most collisions occur between April and May
- For **automobiles**, **pedestrians**, and **public transit vehicles**, collisions are at a yearly high towards the extremes of the year



Modeling Results: R^2

The first model, `overall_model`, was run using neighbourhood, hour, day-of-week, month, and vehicle type as predictors, with number of collisions as the outcome variable:

```
1 overall_model <- lm(number_of_collisions ~ neighbourhood + occ_month + occ_dow +  
2                       occ_hour + vehicle_type, data = overall_collisions)
```

- The R^2 value for this model is **0.3241**.

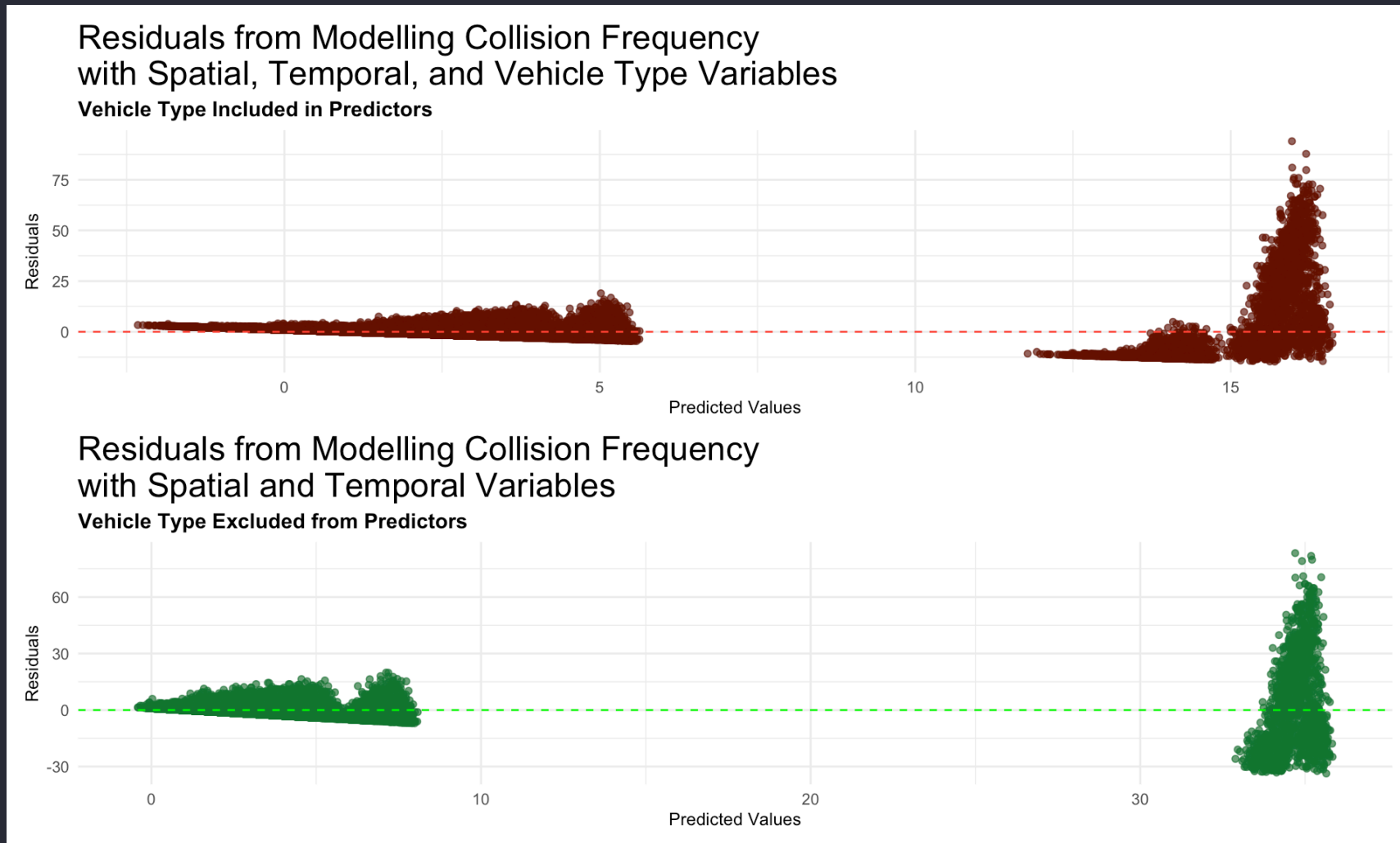
The second model, `minus_vt_model`, was run to investigate if model performance would improve if vehicle type were removed as a predictor:

```
1 minus_vt_model <- lm(number_of_collisions ~ neighbourhood + occ_month + occ_dow  
2                       + occ_hour, data = overall_collisions_minus_vt)
```

- The R^2 value for this model is **0.5806**.

Modeling Results: Residuals

To evaluate the models and how well they fit the data side-by-side, the residual errors from both models were plotted versus their respective predictions.



Discussion

Findings: Exploratory Data Analysis

- We can temporally and spatially model collision risk, similar to prior research
 - Data science on collision risk can help prevention measure development
- We do see variation in collision risk trends for different vehicle types
- All modes of transportation experience sharp increases in collision frequencies during the morning rush hour, and hit their peaks during the afternoon rush hour
- More accidents occur during the week, when people are commuting
- More accidents for year-round transit modes like automobiles and public transit vehicles happen during the winter
 - Bicycles and motorcycles, which are used less frequently during the winter, record far more collisions over the summer

Findings: Modelling

- While spatial and temporal trends can be observed visually, they are not particularly well-correlated
 - Residuals are centered around zero, but have observable patterns and a non-constant spread
 - Removing vehicle type to the model nearly doubled the R^2
- More data on non-automobile collisions might be necessary to properly use vehicle type as a predictor

Limitations

- Predominantly automobile collisions recorded
 - Over 350,000 automobile collisions were recorded, while public transit vehicles were under 50,000, and other modes of transportation did not even break 10,000
 - Even if vehicle type were an accurate predictor of accident frequency, fitting a model with vehicle type as a predictor would be challenging
- Using data collected by a police service
 - Only collisions severe enough to warrant the police responding to the scene are recorded
 - Involved parties might have any number of reasons for not wanting to involve the police

Conclusion

Key Takeaways

- Traffic collisions in Toronto exhibit clear temporal and seasonal patterns, with risks varying by travel mode.
- Recent TPS data can effectively identify collision trends and high-risk periods.

Future Research

- Incorporate additional variables such as weather and land use to improve explanatory power.
- Explore more advanced spatial and spatiotemporal models to better capture collision hotspots.
- Extend the analysis with additional years of data to evaluate long-term trends and the impact of future transportation policies.

References

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